

4. Set the IPv6 source to an operational [IPv6 Unicast Address](#) of the sending Node.
5. Set the IPv6 UDP port number to the [Matter IPv6 multicast port](#).

4.16.3. Receiving a group message

All Nodes supporting groups SHALL register to receive on the associated [IPv6 multicast address](#), at the [Matter IPv6 multicast port](#), for each group of which they are a member.

Upon receiving an IPv6 message addressed to one of these Multicast Addresses the Node is registered for, the Node SHALL:

1. Extract the message from the IPv6 payload.
2. Perform [Section 4.7.2, “Message Reception”](#) processing steps on the message.

4.17. Group Key Management

This section describes operational group keys, a mechanism for generating, disseminating and managing shared symmetric keys across a group of Nodes in a Fabric. Operational group keys are made available to applications for the purpose of authenticating peers, securing group communications, and encrypting data. These keys allow Nodes to:

- Prove to each other that they are members of the associated group
- Exchange messages confidentially, and without fear of manipulation by non-members of the group
- Encrypt data in such a way that it can only be decrypted by other members of the group

A central feature of operational group keys is the ability to limit access to keys to a trusted set of Nodes. In particular, credentials required to generate operational group keys SHALL only be accessible to Nodes with a certain level of privilege — those deemed a member of the group. Barring software error or compromise of a privileged Node, access to shared keys SHALL be computationally infeasible for non-trusted parties.

Operational group keys are shareable across all types and combinations of Nodes as determined by the Administrator managing the group. Given all Nodes in possession of the current epoch keys for the group can communicate with other Nodes in the group, it is the responsibility of the Administrator managing the group to only compose groups of Nodes where communication is appropriate for the given application and security requirements.

4.17.1. Operational Groups

An operational group is a logical collection of Nodes that are running one or more common application clusters and share a common security domain in the form of a shared, symmetric group key. For example, a set of Nodes running a lighting application can form an operational group by sharing a common operational group key derived from the mechanisms described here. Subgroups can be formed within the operational group by defining distinct Group Identifiers for each set of Nodes, while sharing a common operational group key.